SELLINGHIGHS RAIN MARKETS

Data Edge & Developer Build Brief

Goal: build a proprietary rain-market intelligence layer that estimates the exact precipitation amount the Kalshi settlement station will ultimately report - not just whether a city is likely to get rain.

Core thesis

There are two independent opportunities for edge: (1) meteorological edge - how much precipitation physically falls at the exact settlement point; and (2) measurement/settlement edge - how that specific station, gauge, equipment era, and reporting pipeline converts the weather into the value Kalshi settles against.

Primary product concept: a station-specific "True Rain" estimate plus proprietary signals for station bias, upstream rain, and settlement delta.

1. What We Are Actually Predicting

The system should treat each Kalshi rain market as a prediction problem tied to one specific settlement station and one specific settlement product. The target is the eventual reported precipitation total at that station, after observation, gauge behavior, quality control, reporting lags, and final climate-product reconciliation.

Do not build a generic city rain forecast

A normal user sees "Chicago rain forecast." SellingHighs should model "What will the exact Chicago settlement station report in the final settlement source?"

Two-layer model

Layer	Question	Examples of edge
Meteorological	How much precipitation reaches the exact station coordinate?	Radar nowcast, upstream gauges, convective uncertainty, virga detection, ensemble distributions
Measurement / settlement	How much will the station and final reporting pipeline record?	Gauge bias, wind undercatch, equipment era, high-rate error, reporting lag, manual/final corrections

2. Highest-Value Data & Signals to Build

Signal / data source	Why it matters	Priority
Exact settlement-station gauge/equipment type	Different precipitation sensors have different behavior, limits, and failure modes.	P0
Station-specific historical bias model	Learns how the exact gauge reports versus independent rainfall estimates under different conditions.	P0
MRMS precipitation at exact station pixel	High-resolution point estimate is more useful than a city forecast.	P0
Nearby + upstream gauge triangulation	Shows whether the settlement gauge is locally wet/dry and can act as a leading indicator.	P0
Radar storm-motion nowcast	Projects rain cells across the exact station coordinate over the next minutes/hours.	P0
Current/unsettled ASOS precipitation	Bridges the gap between official-to-date amount and rain that has already occurred but is not finalized.	P0
Convective vs stratiform classification	Point uncertainty is dramatically larger in thunderstorms than broad stratiform rain.	P1
Wind + gauge catch relationship	Strong wind may create systematic undercatch; direction may expose local micro-site effects.	P1
Equipment/maintenance era history	Historical station behavior can change after gauge replacement, station moves, or maintenance.	P1
Ensemble/model distribution at station coordinate	Produces a probability distribution and market probability, not a single deterministic forecast.	P1
Virga / ground-reach probability	Prevents radar echoes aloft from being mistaken for measurable surface rain.	P1
Dual-pol radar features	Helps distinguish hydrometeors, hail contamination, melting layer, and heavy-rain regimes.	P2
Settlement pipeline reconciliation	Models how provisional values become the final climate-report value used for settlement.	P1

3. Build a "Rain Gauge Personality" for Every Kalshi Station

Each settlement station should have a persistent station profile built from years of observations and independent estimates. The profile should answer: under this exact weather regime, does this station tend to report wet, dry, or normally?

- Station identifier and exact latitude/longitude.
- Settlement market(s) and final settlement source.
- Gauge/sensor type, manufacturer/model where available, install/replacement dates.
- Station relocation, maintenance, and equipment-change history.
- Historical official precipitation totals.
- MRMS QPE at the station pixel and surrounding pixels.
- Nearby ASOS/AWOS, mesonet, HADS/MADIS, and CoCoRaHS gauges.
- Wind speed/direction during rainfall.
- Rain rate/intensity.

- Storm regime: stratiform, mixed, convective, tropical, winter/mixed precipitation.
- Humidity/dew point and other indicators of evaporation/virga.

Example learned station behavior

```
KXYZ station profile

Light rain:                  median bias -0.01 in
Moderate stratiform:         median bias  0.00 in
Heavy convection:            capture ratio 0.95
Wind > 25 kt:                capture ratio 0.93
Very high rainfall rate:     capture ratio 0.91
Current equipment era:       2023-present
```

4. MRMS at the Exact Gauge Pixel

Ingest NOAA Multi-Radar/Multi-Sensor precipitation products at the exact settlement coordinate. Retain both the station pixel and local neighborhoods (for example 5 km and 10 km averages) so the system can distinguish a true point miss from a station/gauge divergence.

```
Example live panel

MRMS station pixel:      0.37 in
MRMS 5 km mean:          0.41 in
MRMS 10 km mean:         0.46 in
Settlement gauge:        0.31 in
Current gauge/MRMS:      0.84
Expected regime ratio:   0.93
```

5. Proprietary Signal: Settlement Gauge Bias Index

Continuously compare the settlement gauge with MRMS and nearby gauges, conditioned on the current storm regime. The key is not raw divergence alone; it is divergence relative to what is normal for this station under similar rain rate, wind, and storm type.

```
CURRENT GAUGE BIAS

MRMS estimate:           1.13 in
Observed gauge:          1.00 in
Raw divergence:         -11.5%
Expected divergence:     -3.2%
Abnormal station bias:   -8.3%
Confidence:              HIGH
```

6. Nearby Gauge Network + Upstream Rain

Build rings of nearby gauges, but weight them by storm motion instead of distance alone. A gauge southwest of a station may be much more informative than a closer gauge east of it if storms are moving southwest-to-northeast.

- 0-2 km ring
- 2-5 km ring
- 5-10 km ring
- 10-25 km ring

Proprietary Signal: Upstream Rain

```
Storm motion: 237 deg -> 057 deg
Speed: 31 mph

UPSTREAM GAUGES
Gauge A   12.4 mi SW   ETA +24 min   0.18 in / 15m
Gauge B    8.1 mi SW   ETA +16 min   0.21 in / 15m
Gauge C    4.2 mi SW   ETA  +8 min   0.17 in / 15m

Projected station arrival: 12:42 PM
Projected station increment: 0.14-0.24 in
```

7. Treat Convective Rain as a Different Prediction Problem

The same model mean should not imply the same confidence under all storm types. Broad stratiform rain is relatively spatially smooth. Thunderstorms can produce huge rainfall gradients across an airport. The distribution around the exact station should widen substantially for convective events.

- STRATIFORM
- MIXED
- CONVECTIVE
- TROPICAL
- WINTER / MIXED PRECIPITATION

8. Model the Gauge Itself

Use equipment metadata and station history to divide backtests into equipment eras. Historical behavior before a sensor replacement or station relocation should not automatically be treated as the same population as the current era.

Rain-rate-dependent error

Empirically learn whether the gauge capture ratio changes as instantaneous rain rate increases. For mechanical/tipping systems, the highest-intensity events may behave differently than light rain.

```
capture_ratio = f(rain_rate, wind_speed, wind_direction, gauge_type, temperature, storm_type, equipment_era)
```

Wind-induced undercatch / micro-site effects

Condition station-vs-independent-estimate error on wind speed and wind direction. Direction matters because buildings, runway structures, trees, sensor placement, and turbulence can make local aerodynamics asymmetric.

9. Learn the Reporting & Settlement Pipeline

Do not assume the latest visible rainfall number is already the final settlement value. Reconstruct how raw/incremental station precipitation becomes hourly/daily climate values and ultimately the final settlement report. Backtest where provisional values differ from final climate-product values.

```
Raw / incremental station observations
          |
Hourly precipitation / METAR remarks
          |
Daily accumulation / climate report
          |
Monthly accumulation / final report
          |
Kalshi settlement value
```

- How often does the provisional cumulative total differ from the eventual final value?
- Which stations receive manual corrections most often?
- How are missing observations backfilled?
- How are trace amounts handled in daily and monthly accumulation?
- How long is the typical reporting lag after a storm?
- Does the settlement source use preliminary or finalized data for the specific market?

Proprietary Signal: Settlement Delta

Settlement Delta should estimate the difference between the currently visible official amount and the amount SellingHighs expects the final settlement report to contain after already-observed but unsettled rain and likely corrections are incorporated.

10. Virga / Ground-Reach Probability

Radar echo does not guarantee measurable rain reaches the gauge. Use humidity, temperature-dew-point spread, cloud base, sounding/model humidity, and radar characteristics to estimate whether precipitation evaporates before reaching the surface.

```
Radar echo overhead:        24 dBZ
Surface RH:                 31%
Temperature-dew point gap:  large
Cloud base:                 high
Ground-reach probability:   LOW
Expected measurable rain:   0.00-0.01 in
```

11. Dual-Polarization Radar (Advanced)

- Reflectivity
- Differential reflectivity
- Correlation coefficient
- Specific differential phase

These inputs can improve hydrometeor classification and rainfall-rate estimation, especially in hail, melting-layer, tropical, or high-intensity events. This is an advanced enhancement rather than the first MVP requirement.

12. Future Rainfall: Model the Distribution, Not a Single Forecast

For the remaining market period, extract numerical-model and ensemble guidance at the exact settlement-station coordinate. Retain members/percentiles instead of averaging everything into one deterministic amount.

- HRRR / rapid-refresh guidance for near term
- High-resolution ensemble guidance where available
- Global ensembles farther out
- WPC QPF as an additional benchmark
- Commercial sources only if licensing/value justify them

```
Remaining-period rainfall distribution

P10       0.31 in
P25       0.52 in
Median    0.79 in
```

```
P75       1.23 in
P90       1.84 in
```

Forecast output should be strike probabilities

Combine current official accumulation, already-observed-but-unsettled precipitation, near-term nowcast, station bias adjustment, and future ensemble distribution. Convert the result directly into the probability that final settlement finishes above/below each Kalshi strike.

13. Main Product Feature: "True Rain"

True Rain should be the proprietary SellingHighs estimate of where the settlement station effectively stands right now, even if official climate reporting has not caught up.

```
KMDW TRUE RAIN

Official settled-to-date:        6.42 in
Unsettled station accumulation:  +0.21 in
MRMS / gauge adjustment:         -0.02 in
---------------------------------------
TRUE RAIN NOW:                    6.61 in

Expected remainder:              +1.17 in
---------------------------------------
Projected final settlement:       7.78 in
80% range:                        7.12-8.91 in
```

Supporting station signals

```
Station Bias        -2.8%
Radar Confidence      91%
Gauge Health          Normal
Storm Type            Convective
Upstream Rain         Heavy
Model Agreement       Medium
Settlement Delta      +0.18 in
```

14. Suggested Data Architecture

- station_registry - settlement station, coordinates, market mappings, settlement source, timezone.
- station_equipment - gauge type, model, install/remove dates, equipment era, maintenance notes.
- official_precip - observed official rainfall by station/time, provisional/final flags.
- mrms_precip - station-pixel and neighborhood QPE/rain-rate features.
- nearby_gauges - observations from eligible gauge networks, coordinates, quality flags.
- radar_features - storm motion, reflectivity/dual-pol fields, cell tracking, projected ETA to station.
- weather_features - wind, direction, humidity, dew point, temperature, cloud base, regime classification.
- forecast_members - per-model/per-member rainfall at the station coordinate by valid time.
- settlement_history - final Kalshi settlement source/value and any provisional-to-final changes.
- derived_signals - True Rain, Station Bias, Upstream Rain, Settlement Delta, confidence.

Core processing pipeline

```
Collectors -> normalized time-series store -> QC -> station feature builder
          -> live nowcast / bias models -> settlement distribution
          -> Kalshi strike probabilities -> SellingHighs UI + alerts
```

15. Modeling & Backtesting Requirements

- Train and score by settlement station, not only globally.
- Use equipment era as a feature or hard backtest partition.
- Prevent look-ahead leakage: only use data that would have been available at the historical market timestamp.
- Measure calibration of probabilities, not only MAE of rainfall totals.
- Backtest threshold/strike accuracy and simulated market edge after fees/slippage assumptions.
- Record model performance separately for stratiform vs convective regimes.
- Track whether edge comes from weather forecasting, station measurement bias, or settlement/reporting latency.
- Generate confidence intervals and suppress/discount signals when data quality is poor.

16. Recommended MVP Order

Phase 1 - Settlement station registry

- Map every supported rain market to the exact station, coordinates, settlement source, timezone, and gauge/equipment metadata.
- Create equipment-era history where possible.

Phase 2 - True Rain foundation

- Ingest official precipitation plus MRMS at the exact station pixel.
- Compute official vs MRMS divergence and basic current True Rain.

Phase 3 - Nearby/upstream intelligence

- Add nearby gauges, storm-motion vectors, ETA weighting, and Upstream Rain.
- Add simple stratiform/convective classification.

Phase 4 - Station Bias model

- Train station-specific capture/bias models conditioned on rain rate, wind, storm type, and equipment era.
- Create Gauge Bias Index + confidence.

Phase 5 - Settlement Delta

- Reconcile provisional station data against final settlement/climate reports.
- Estimate already-observed but not-yet-finalized precipitation and correction risk.

Phase 6 - Future distribution + market probabilities

- Ingest ensemble/model members at exact station coordinates.
- Combine current True Rain + future rainfall distribution to compute probability for each strike.

Phase 7 - Advanced radar physics

- Virga detection, dual-pol features, hydrometeor classification, higher-end cell tracking.

17. Suggested SellingHighs Rain Page UI

- Top card: market/station name, settlement source, current official amount, True Rain, projected final settlement, probability range.
- Station card: gauge type, equipment era, Station Bias, Gauge Health, current wind and storm regime.
- Live radar card: MRMS station pixel, neighborhood mean, current rain rate, radar confidence.
- Upstream Rain card: storm vector, upstream gauges/cells, ETAs, projected increment.
- Settlement card: official-to-date, unsettled observed rain, estimated correction/lag, Settlement Delta.
- Forecast distribution card: P10/P25/median/P75/P90 remaining rain and final-settlement distribution.
- Market table: every Kalshi strike with market-implied probability, SellingHighs probability, edge, confidence, and last-update timestamp.
- History tab: past station bias by rain-rate/wind/storm regime plus equipment changes.

18. Proprietary SellingHighs Signals

Signal	Definition	Why useful
True Rain	Best estimate of effective current settlement accumulation after unsettled observations and measurement adjustments.	More actionable than stale official-to-date totals.
Station Bias	Expected station reporting bias versus independent precipitation estimates under the current regime.	Turns station quirks into a repeatable feature.
Upstream Rain	Rain currently upstream of the station, projected by motion and arrival time.	Gives a near-term leading indicator before it hits the gauge.
Settlement Delta	Expected difference between latest visible official total and eventual finalized settlement total from already-observed precipitation/corrections.	Targets reporting-lag and reconciliation edge.
Gauge Health	Confidence that the station is behaving normally based on nearby/radar consistency and data quality.	Prevents overtrusting anomalous or missing station data.

19. First Concrete Developer Milestone

Build one station end-to-end

Pick one active Kalshi rain settlement station. Ingest its official rain observations, exact coordinates, gauge/equipment metadata, MRMS station-pixel precipitation, a set of nearby gauges, wind/humidity, and one future model/ensemble source. Show Official Rain, True Rain, Station Bias, Upstream Rain, Settlement Delta, projected final rainfall distribution, and strike probabilities on one live page. Backfill enough history to test whether the station-bias and settlement-delta signals have predictive value.

20. Success Criteria

- Every rain market is mapped to the correct station and settlement source.
- All time series are normalized to station-local time and UTC internally.
- Live values identify source, observation time, ingestion time, and provisional/final status.
- Historical backtests are point-in-time safe with no future-data leakage.
- True Rain can explain every adjustment from the official visible number.
- Station Bias includes a confidence score and regime/equipment context.
- Upstream Rain produces ETA and expected increment at the station.
- The system outputs a full probability distribution, not only a single total.
- SellingHighs probability and market-implied probability can be compared strike-by-strike.
- Data-quality failures degrade confidence rather than silently producing a precise-looking signal.

Strategic focus: do not try to beat every weather forecast everywhere. Become exceptionally good at the small set of exact physical sensors and reporting pipelines that settle the markets you trade.